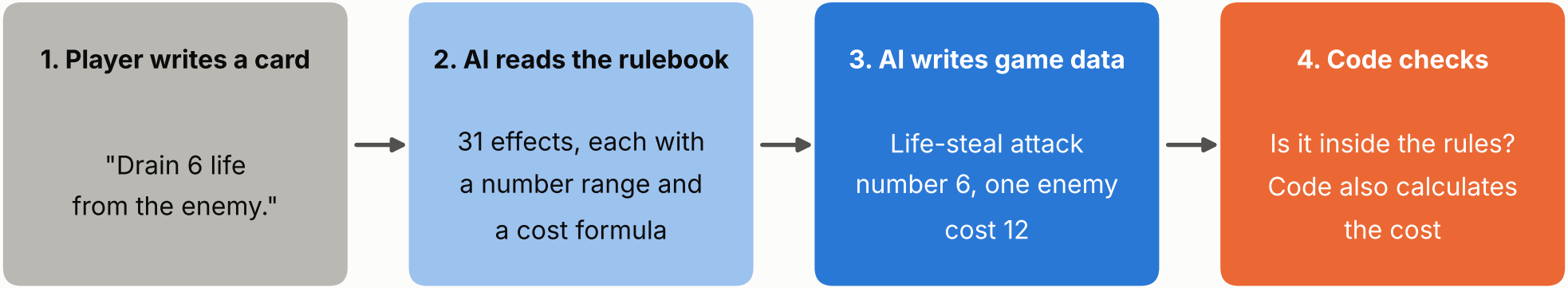
The card test: can the AI follow the game's rulebook?

In my game idea, players write their own cards in plain English. The AI turns each card into game data.



Rulebook groups: attack · shield · healing · drawing cards · energy · status effects (burn, chill, poison, weaken)

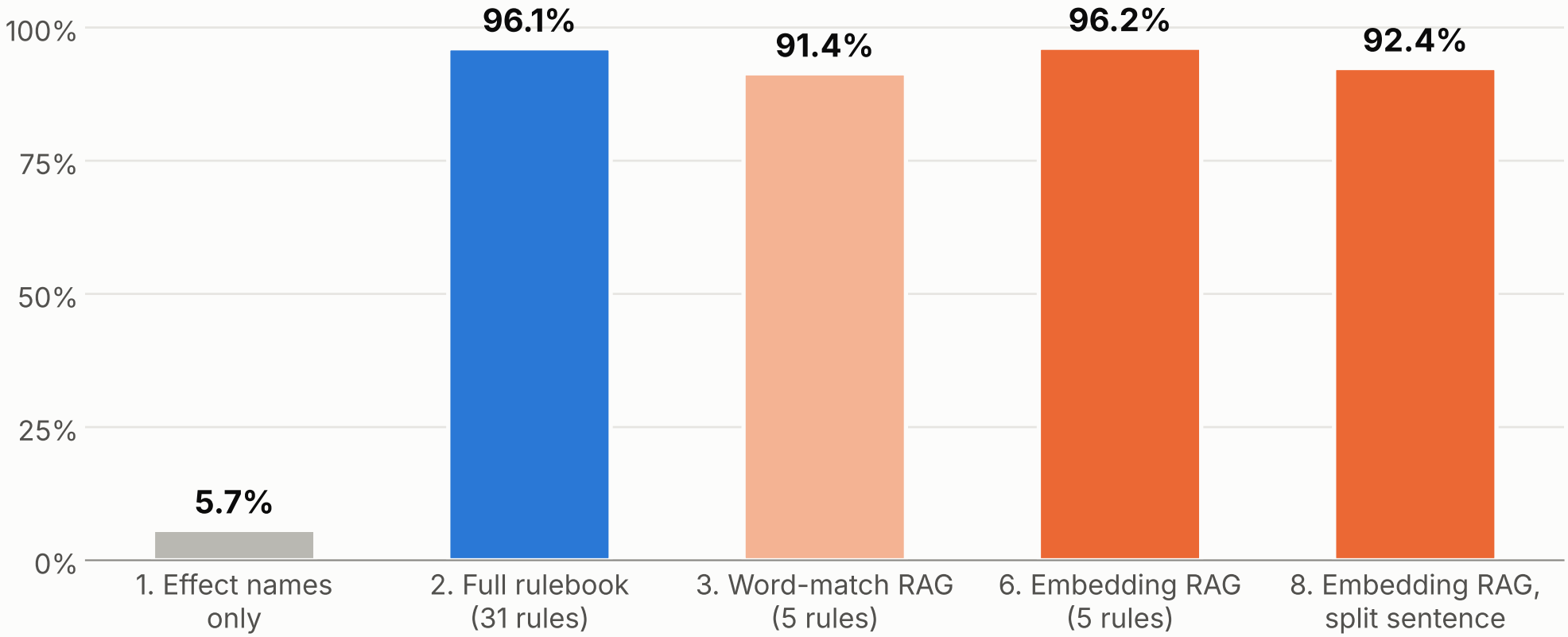
Some effects look alike on purpose, e.g. a shield that lasts one turn vs a permanent shield that costs twice as much.

Test: 35 cards, correct answers decided in advance, graded by code. Some must be rejected ("Deal 40 damage": max is 30).

RAG = the AI gets only the 5 rules the search thinks are most related to the card, not the whole rulebook.

Embedding RAG matched the full rulebook

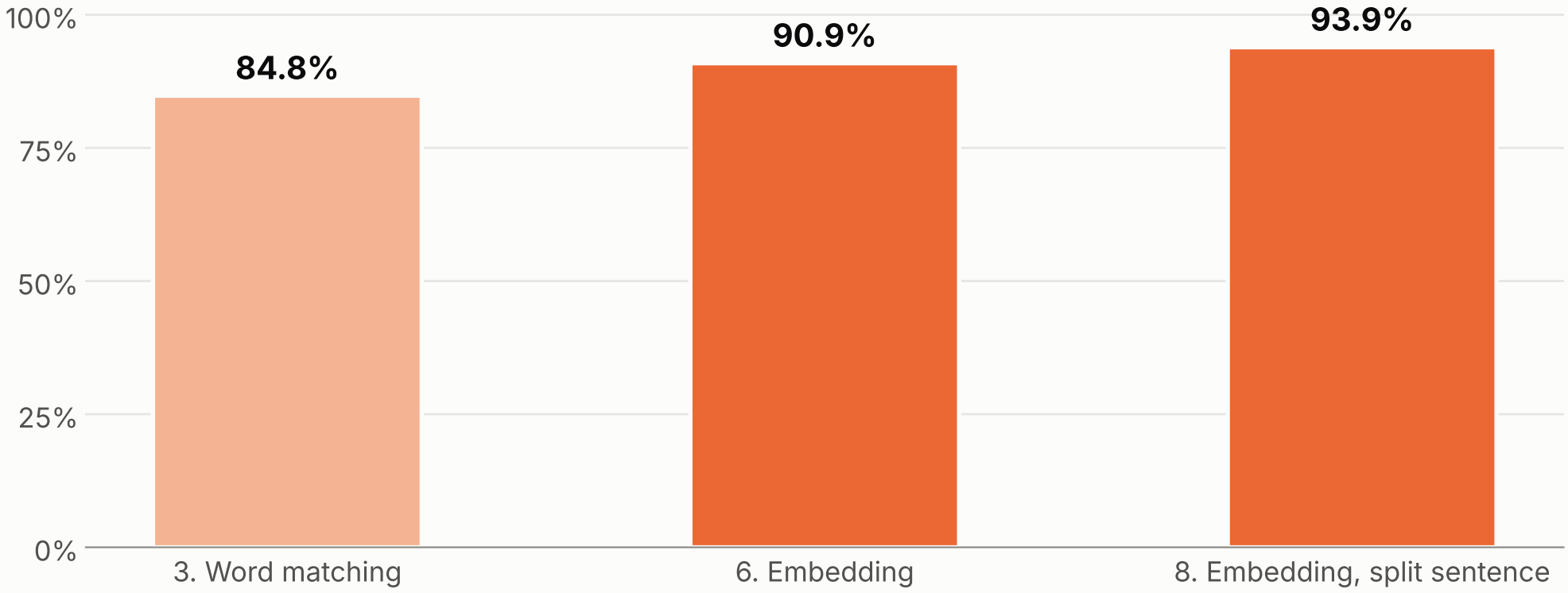
35 player-written cards × 3 runs. Correct = right effects and right cost, or a correct rejection.



Embedding model: gemini-embedding-001. Versions 3, 6, 8: same prompt, only the rule search is different.

Searching by meaning found more of the needed rules

Cards where every needed rule was among the 5 picked rules (33 cards that need a rule).

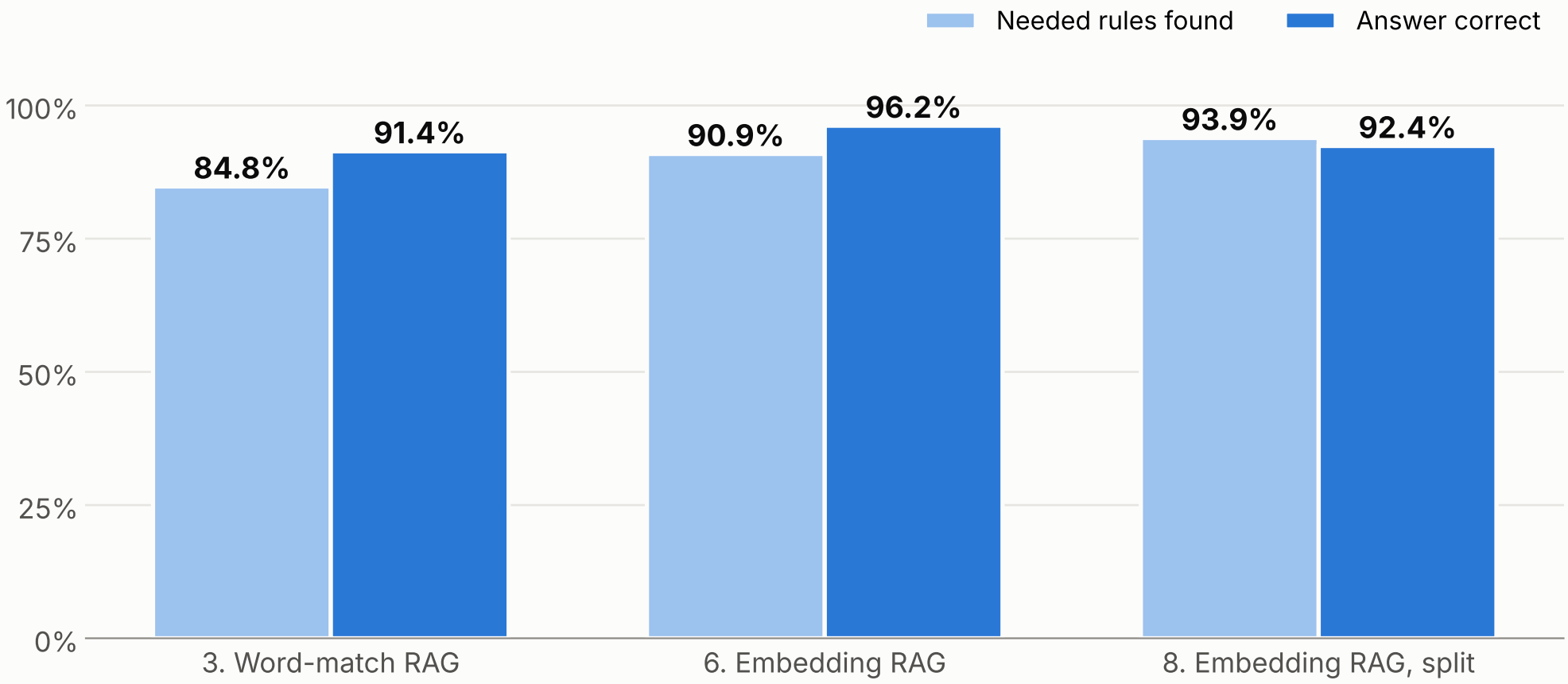


Fixed by meaning search: "Drain 6 life from the enemy" now finds the life-steal attack rule.

Fixed by splitting: "Deal 4 damage and heal 6" finds both rules. Newly missed: "...gain 12 shield".

Finding more rules did not always mean more correct answers

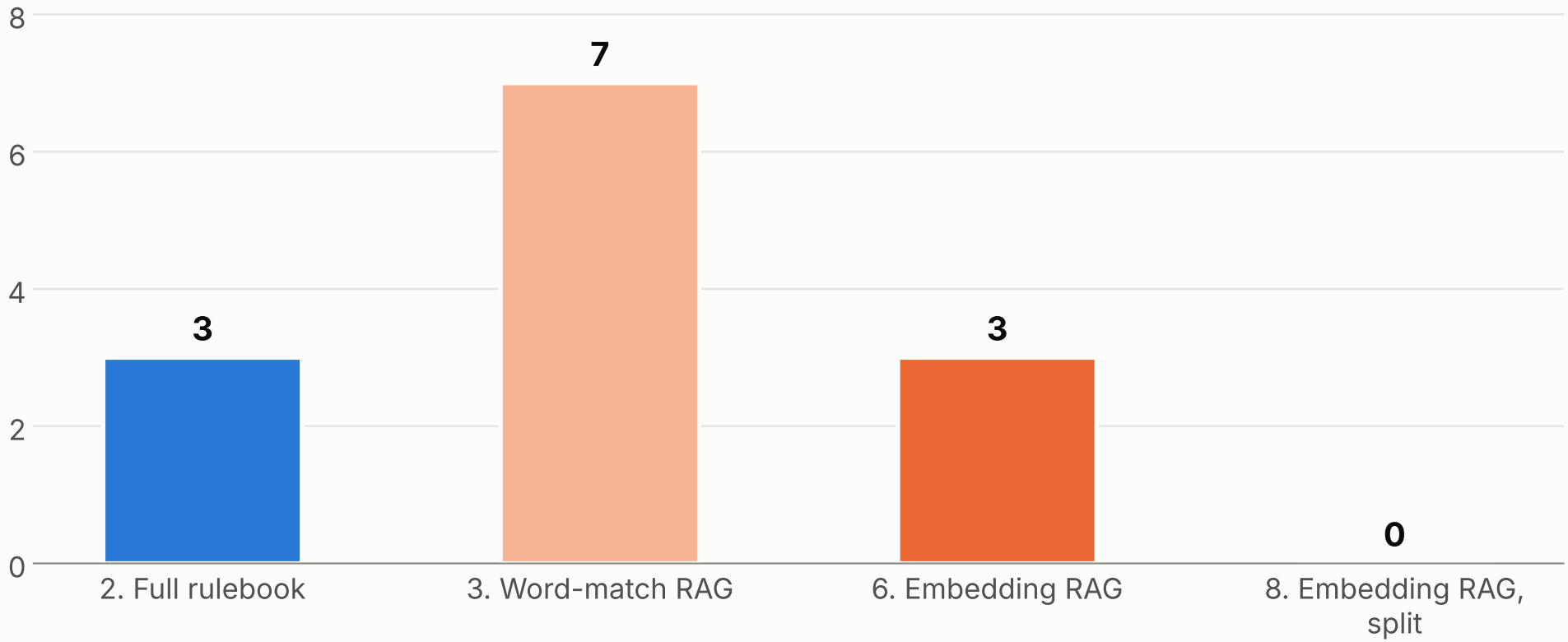
Same 35 cards. Light = needed rules found. Dark = final answer correct.



"Below half health, gain 12 shield": the normal shield rule was missed, so the AI used the permanent one (2× cost).
"Deal 6, and 6 more if burning": an extra, unneeded attack rule made the AI combine two effects wrongly.

Valid cards that were wrongly rejected

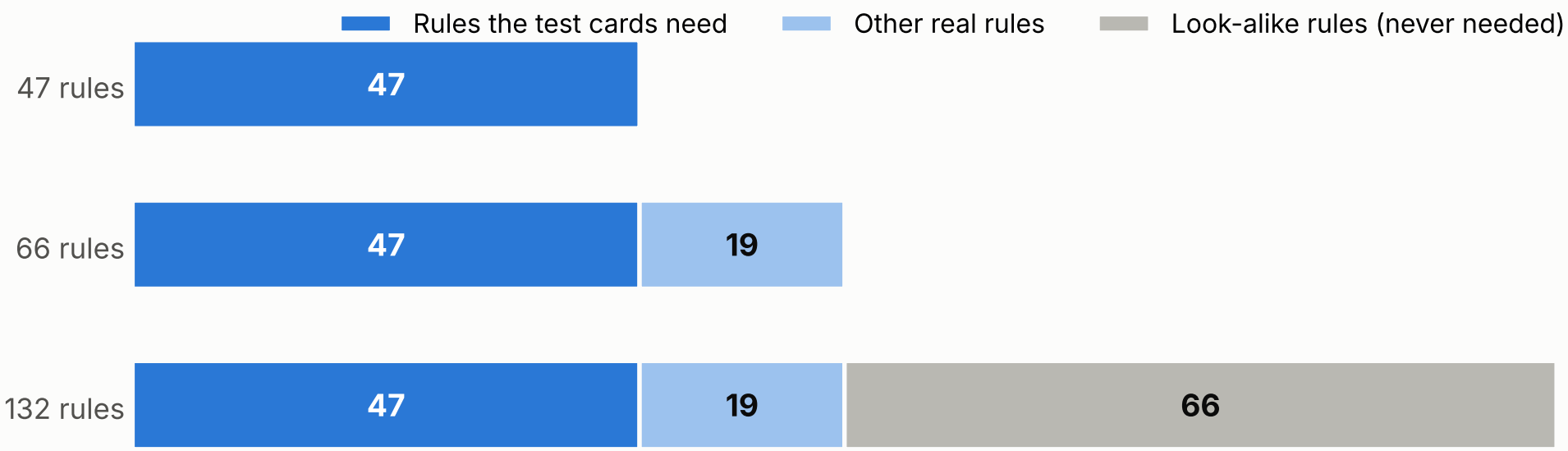
Out of 84 tries per version (28 valid cards × 3 runs).



The code check (versions 4, 5, 7) stepped in 0 times again. The 31-rule rulebook is too easy for the AI to break a rule.

Next: a bigger rulebook with 3 difficulty tiers

Same 66 test cards for every size. Only the rulebook grows.



Tiers: 1 = basic (22 rules) 2 = status effects and look-alikes (26) 3 = special rules (18)

Tier 3 examples: "cannot be on the same card as an attack", "must be the last effect", "cost not doubled".

Test cards: 66 (Tier 1: 20, Tier 2: 20, Tier 3: 26). 48 should pass, 18 should be rejected. All answers and costs checked by code.

Versions per size: full rulebook / word-match RAG / embedding RAG / embedding RAG + code check.